

Lillian Miller

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EDUCATION

University of Pennsylvania, College of Arts and Sciences, Philadelphia, PA

May 2029

Candidate, Doctor of Philosophy in Earth and Environmental Sciences

GPA: 4.0/4.0

Honors & Awards: National Science Foundation (NSF) Graduate Research Fellowship (GRFP), Penn Arts and Sciences Graduate Student Government Representative

University of Pennsylvania, College of Arts and Sciences, Philadelphia, PA

May 2025

Bachelor of Arts, Chemistry & Earth and Environmental Science

GPA: 3.97/4.0

Honors & Awards: Vagelos Molecular Life Sciences Scholar, Phi Beta Kappa, Vagelos Science Challenge Award, Rose Undergraduate Research Award, Marshall Scholarship Finalist, ACS Undergraduate Analytical Chemistry Award, Joullié Scholar

RESEARCH EXPERIENCE

Climate Dynamics Lab, University of Pennsylvania | *Graduate Researcher (Undergraduate Researcher, 2023–2025),*

Philadelphia, PA — Advisor: Dr. Irina Marinov

January 2023 – Present

- Analyzing model, satellite, and in situ datasets in Python (xarray, pandas, visualization libraries) on high-performance computing platforms, porting workflows between Penn's General-Purpose Cluster and the NCAR supercomputer.
- Synthesize literature across project areas, including ecological trends at Ocean Station PAPA, ocean carbon chemistry and buffering systems, and marine carbon dioxide removal technologies; distilled a review of 18 CMIP6 models with marine biogeochemistry components into 40 pages of fact sheets and a searchable spreadsheet used lab-wide.
- Present research to scientific and general audiences: international conference posters (Ocean Sciences Meeting, Glasgow), campus outreach events, and summer research symposia.
- Adapted research code into course materials; mentoring multiple undergraduate researchers in project design, literature and code review skills, and research presentation.

Projects:

- Characterizing the carbon fingerprints of deep convection events and open-ocean polynyas in the Southern Ocean across CMIP6 models, isolating their signature on air–sea carbon exchange.
- Identifying interannual biogeochemical variability across the Southern Ocean — phytoplankton phenology, nutrient co-limitation — from satellite composites and ARGO float data, and assessing likely responses to climate change.
- Utilized PyCO2SYS software in python to understand and run the ocean carbon chemistry subroutines utilized in climate models and evaluate carbonate chemistry across the Southern Ocean
- Investigated ecological drivers of phytoplankton productivity in the iron- and light-limited North Pacific and North Atlantic subpolar regions, evaluating CMIP6 models against observational products (World Ocean Atlas, satellite chlorophyll-a and phytoplankton biomass products); co-author on manuscript in preparation

TEACHING & COMMUNICATION EXPERIENCE

Penn Earth and Environmental Science Department | *Teaching Assistant*, Philadelphia, PA January – May 2025, 2026

- Taught and mentored 25+ students in Python-based large scale data analysis and visualization projects
- Managed organizational aspects of the course including coordinating deadlines, maintaining course materials, and ensuring streamlined communication between the students and instructional team

Penn Climate | *Student Advisor*, Philadelphia, PA

August 2024 – May 2026

- Member of the student advisory committee focused on increasing student engagement in environmental issues
- Organized 2025 Climate Week marine heat wave outreach event; and collaborated with leaders of Penn's climate faculty working group to create a climate “venting space” event. Coordinated similar events for Penn's 2026 Earth Week.

Netter Center | *Environmental Education Coordinator*, Philadelphia, PA

April - August 2025

- Designed and implemented hands-on environmental curriculum for Philadelphia high school students
- Led field experiences and guest speaker events highlighting climate and sustainability careers

Perkiomen Watershed Conservancy | *Environmental Educator*, Schwenksville, PA

May – August 2022

- Designed and led interactive field programs for 100+ K–12 students, teaching water quality, plastic pollution, and stream ecology
- Operated and maintained a floating classroom for hands-on environmental education, ensuring safety and engagement

POLICY EXPERIENCE**Perry World House** | *Incoming Graduate Fellow*, Philadelphia, PA

September 2026 – Present

- Selective fellowship training graduate researchers to translate scientific research into actionable global policy.

American Geophysical Union (AGU) | *Local Science Partner*, Philadelphia, PA

January 2026 – Present

- Competitively selected to bridge the scientific community and policymakers at local and federal levels, and develop skills to communicate complex science and environmental policy issues to legislators and public audiences
- Met with federal policymakers in Washington, D.C. to advocate for evidence-based environmental policy and science funding in April 2026

Penn Science Policy & Diplomacy Group

February 2026 – Present

- Working with a team to write a policy memo for Pennsylvania state policy makers on PFAS

Kleinman Center Climate & Energy Policy Seminar | *Undergraduate Student Fellow*, Philadelphia, PA

January – May 2024

- Actively participated in bi-weekly seminars and networking sessions, learned about various topics in climate and energy policy including carbon pricing, energy politics in the EU, and energy efficient design from experts in these fields.
- Researched and wrote an article published on the Kleinman Center website on the importance of data science in solving the climate crisis, with the example of utilizing satellite data in crafting policy and management strategies to prevent methane leakages.

SELECTED HONORS & AWARDS**National Science Foundation (NSF) Graduate Research Fellowship (GRFP)**

May 2026

- Competitively awarded to 2,500 of nearly 14,000 applicants nationwide; provides three years of funding over five years, including a \$37,000 annual stipend and \$16,000 cost-of-education allowance.

Roy and Diana Vagelos Science Challenge Award

June 2024

- One of ~five students selected annually, for academic merit and research commitment; covered full tuition and fees for 2024-25.

Rose Undergraduate Research Award

May 2025

- Recognizes ten graduating seniors each year for outstanding research projects, with a reward of \$1000

Vagelos Molecular Life Sciences (MLS) Scholar

August 2021– May 2025

- Undergraduate program that emphasized developing fundamental knowledge in mathematics, chemistry, and physics.
- One of approximately 50 undergraduates selected to participate upon admission to the University of Pennsylvania. One of 25 students in my year to meet criteria to continue with the program and receive funding for research endeavors.

SELECTED PRESENTATIONS & PUBLICATIONS

- Marinov I., **Miller L.**, Tran J., Cabre A., Sharma P. Ocean phytoplankton ecology in the North-East Pacific and North Atlantic subpolar regions across recent generations of climate models. (In Progress)
- **Miller, L.**, Marinov, I., Cabre, A., Tran J., Sharma, P., (2026, February) *Ocean Phytoplankton Ecology and Growth Limitations: Northern Subpolar Lessons from Observations and CMIP6 Models* [Poster], Ocean Sciences Meeting, Glasgow, Scotland
- Lu, S., **Miller, L.**, Wang R., Marinov, I., Molodtsov, S. (2026, February), *Heat and carbon exchange signatures of Southern Ocean open-ocean polynyas across CMIP6 Models* [Poster], Ocean Sciences Meeting, Glasgow, Scotland
- Sapkota A. **Miller, L.** Marinov I. (2025, October) *Marine CO₂ Removal: A Review of Current Technology and Challenges* [Poster], University of Pennsylvania Climate Week 2025, Philadelphia
- **Miller L.** 2024 Sep 19. Big Data Applications to Energy and Climate Policy. Kleinman Center for Energy Policy News & Insights. <https://kleinmanenergy.upenn.edu/news-insights/big-data-applications-to-energy-and-climate-policy/>

TECHNICAL SKILLS, COURSEWORK & OTHER INTERESTS

Technical: Python/ Jupyter Notebook, R, MATLAB, SeaDAS software & satellite data processing, High-Performance Computing (HPC) systems, AI-assisted code development (Claude code, GitHub copilot) adobe (illustrator, photoshop)

Relevant Coursework: Satellite & Remote Sensing for Ocean Color (Cornell June 2026), Advanced Statistical Computing, Environmental Law, Environmental Policy, Greenhouse Gas Removals and Net Zero, Risk Analysis and Environmental Management, Energy Systems and Policy

Volunteer Work: Native Plant Project workdays (Perkiomen Watershed)